

Basic CAD Lab

Subject Code 2028 lesson HTML for Mechanical Engineering. This page is a student-friendly study guide with type-specific sections, expected questions and a separate Master Answer Key.

Official syllabus reference: [SITTTR course contents](#) | Author: Nandakumar.M | Visit: nandakumarm.dpdns.org

2028 - Basic CAD Lab

Department(s): Mechanical Engineering. Semester: Semester 2. Subject type: Lab.

2028

Course Code

Basic CAD Lab

2

Semester

Semester 2

Draw

Study Mode

Drawing / Graphics / CADD

How to Study

- Read the overview and make a one-page summary.
- Open each module separately and practise the expected questions.
- Keep diagrams, tables, formulas or procedures neat according to subject type.
- Use the Master Answer Key only after attempting the questions.

Study Support

Use Malayalam for classroom discussion if needed, but keep technical terms in English.
Attempt module questions first, then check the Master Answer Key.

Drawing Bank for Basic CAD Lab

Quick reference cards for this subject type.

Tools / Software

Drawing sheet, pencils, scales, set squares, compass, mini drafter and CAD commands.

Line Types

Object, hidden, center, dimension, projection and cutting-plane lines.

Sheet Layout

Border, title block, view placement, scale, labels and projection symbol.

Construction Steps

Read problem, draw reference, construct lightly, darken final lines and dimension.

Common Mistakes

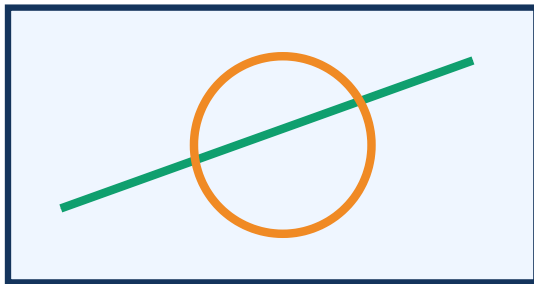
Wrong projection placement, missing center lines, poor dimensioning and untidy lettering.

Drawing Tools and Standards

Use instruments/software tools, line types and sheet conventions.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



Title block

Visual study block for Basic CAD Lab

Simple Explanation

Use instruments/software tools, line types and sheet conventions. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

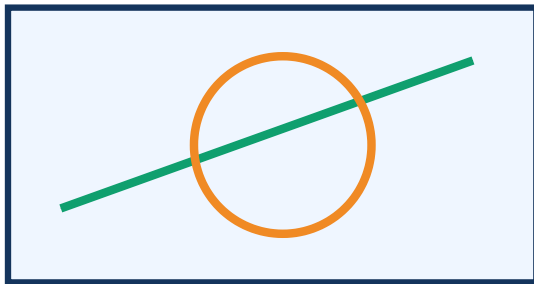
- 1 List drawing tools/software commands used in this subject.
- 2 Differentiate object, hidden, center and dimension lines.
- 3 Write sheet layout rules.
- 4 List common drawing mistakes.

Basic Construction and Dimensioning

Practise construction procedure, scale, lettering and dimensions.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



Title block

Visual study block for Basic CAD Lab

Simple Explanation

Practise construction procedure, scale, lettering and dimensions. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

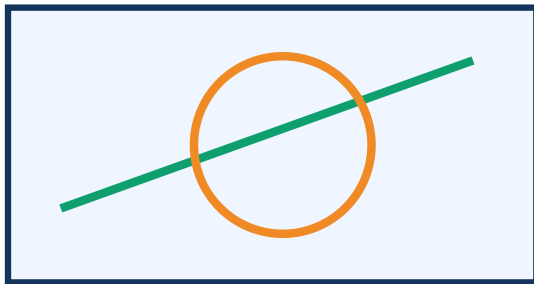
- 1 Write the steps for a neat construction problem.
- 2 Explain dimensioning rules.
- 3 Prepare a checklist for drawing sheet submission.
- 4 State the use of scale and projection lines.

Views, Sections and Details

Develop views, sections, layout and labelled drawing presentation.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



Title block

Visual study block for Basic CAD Lab

Simple Explanation

Develop views, sections, layout and labelled drawing presentation. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

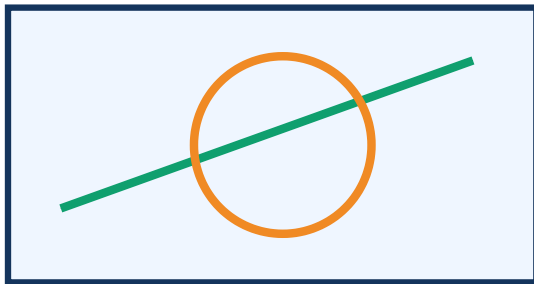
- 1 Explain view placement for a drawing problem.
- 2 State the need for sectional view/detail view.
- 3 Write steps to align multiple views.
- 4 List labelling rules for final drawing.

CAD / Final Sheet Practice

Use CAD/drawing practice to prepare exam-ready sheets.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



Title block

Visual study block for Basic CAD Lab

Simple Explanation

Use CAD/drawing practice to prepare exam-ready sheets. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

- 1 List important CAD commands if applicable.
- 2 Write the final sheet checking method.
- 3 Explain how to avoid overlapping views.
- 4 Prepare a viva checklist for drawing/CAD.

One-page Quick Revision

Last-minute checklist for Basic CAD Lab.

Important Points

- Definitions and key terms.
- Diagrams, formulas, procedures or formats.
- Applications in Mechanical Engineering.

Exam Method

- OK** Read question carefully.
- OK** Write answer in steps.
- OK** Label diagrams and units.
- OK** Finish with result/application.

Common Mistakes

- NO** Skipping units or labels.
- NO** Writing answer without structure.
- NO** Mixing definitions and applications.
- NO** Leaving diagrams untidy.

Answers and Hints

Answer hints exactly match the expected questions inside modules.

Module 1 Answer Hints

M1-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M1-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M1-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M1-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

Module 2 Answer Hints

M2-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M2-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M2-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M2-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

Module 3 Answer Hints

M3-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M3-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M3-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M3-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

Module 4 Answer Hints

M4-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M4-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M4-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

M4-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Basic CAD Lab.

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